

# Cardio MD PM Report

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## *Affected Products*

Cardio MD

## *Reference Material*

PMS-03-XX-001 Cardio MD Service Manual; **IEC 60601-1, Sec 18 Protective Earthing and section 19.2 Single Fault Conditions.**

## *History*

| Date    | Name             | Rev | Change descriptions |
|---------|------------------|-----|---------------------|
| 12/7/14 | Document Control | A   | Initial Release     |
|         |                  |     |                     |
|         |                  |     |                     |
|         |                  |     |                     |

## *Approval*

| Department       | Name               | Date    |
|------------------|--------------------|---------|
| Dept. Manager    |                    |         |
| Document Control | Jehovany Hernandez | 12/9/14 |

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## Preface

This document is valid proof of a Preventative Maintenance performed on the system. The Preventative Maintenance must be performed in the specified intervals, and the results are to be documented herein.

Chapter 5 Maintenance of the Operator's Manual describes user maintenance requirements such as cleaning, check of image quality, simple detector calibrations and checks of safety devices.

This section addresses planned maintenance that should be performed by service personnel. Some of it is also listed in the Operator's Manual for the user to perform on daily and weekly basis.

However, service personnel should make sure that these procedures have been followed and if not, include them on the periodic visits.

**Warning:** All steps must be performed in accordance with the manufacturer's service manual.

This document must be completely filled out by the Field Service Engineer, i.e.:

- All boxes must be filled out. Use **N/A** where appropriate.
- Enter the Serial Number and the Date in the header of each page.
- After completing the check, fill out and sign the last page.

## System Information

|                    |  |
|--------------------|--|
| <b>Customer:</b>   |  |
| <b>Model:</b>      |  |
| <b>System S/N:</b> |  |
| <b>System ID:</b>  |  |

## Measuring Equipment Used

Enter the measuring equipment used in the table below (meters, scopes, phantoms, etc.).

| Equipment | Serial # |
|-----------|----------|
|           |          |
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|           |          |

## Checklist

### One time Maintenance

| Good                     | Bad                      | N/A                      | Action                                                                                              | Frequency                  |
|--------------------------|--------------------------|--------------------------|-----------------------------------------------------------------------------------------------------|----------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Verify that the user is acquainted with the daily/weekly checks described in the Operator's Manual. | 1 month after installation |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Verify that the daily/weekly checks have been performed.                                            | 1 month after installation |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check collision sensors and E-stop.                                                                 | 1 month after installation |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Inspect the image quality, and recalibrate if necessary.                                            | 1 month after installation |

### Every 6 Months

| Good                     | Bad                      | N/A                      | Action                                                                                                                                                                                                                                  | Frequency |
|--------------------------|--------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Verify that the user is acquainted with the daily/weekly checks described in the Operator's Manual.                                                                                                                                     | 6 Months  |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Verify that the daily/weekly checks have been performed.                                                                                                                                                                                | 6 Months  |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check collision sensors and E-stop.                                                                                                                                                                                                     | 6 Months  |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Inspect the image quality, and recalibrate if necessary.                                                                                                                                                                                | 6 Months  |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check Motion limits.                                                                                                                                                                                                                    | 6 Months  |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Grease moving parts.                                                                                                                                                                                                                    | 6 Months  |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Verify that the system ground continuity is intact and in a condition that can conduct a current equal to 25 A without breaking down. <b>Refer to IEC 60601-1, Sec 18 Protective Earthing and section 19.2 Single Fault Conditions.</b> | 6 Months  |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Delete old log files.                                                                                                                                                                                                                   | 6 Months  |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Grease the acme screw inside the line source compartment (If the Cardio MD AC Option is installed).                                                                                                                                     | 6 Months  |

### Uniformity

| Good                     | Bad                      | N/A                      | Action                                                                       | Frequency |
|--------------------------|--------------------------|--------------------------|------------------------------------------------------------------------------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check the uniformity of the detectors by an intrinsic or extrinsic QC check. | Daily     |

### Collision Sensors

| Good                     | Bad                      | N/A                      | Action                                                                             | Frequency |
|--------------------------|--------------------------|--------------------------|------------------------------------------------------------------------------------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check the functionality of all collimator, detector and gearbox collision sensors. | Daily     |

### Hand Controller

| Good                     | Bad                      | N/A                      | Action                                                                                                                 | Frequency |
|--------------------------|--------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check the hand controller for cracks and damage. Check the connection points on the hand controller and on the gantry. | Daily     |

### Table

| Good                     | Bad                      | N/A                      | Action                                          | Frequency |
|--------------------------|--------------------------|--------------------------|-------------------------------------------------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check the table pallet and mattress for damage. | Daily     |

### E-Stop

| Good                     | Bad                      | N/A                      | Action                                                                                                                                                                                                                 | Frequency |
|--------------------------|--------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check both E-Stop buttons for correct operation. When an E-Stop button is activated, no gantry motions should be possible, the yellow E-Stop indicator should be on and the green motor power indicator should be off. | Daily     |

### Cables

| Good                     | Bad                      | N/A                      | Action                                                                                                       | Frequency |
|--------------------------|--------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check cables connected to the PC and monitor. Check for damage to insulation especially around cable clamps. | Weekly    |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check that all external cables are secured in connector clamps where applicable.                             | Weekly    |

### Panels

| Good                     | Bad                      | N/A                      | Action                                                                    | Frequency |
|--------------------------|--------------------------|--------------------------|---------------------------------------------------------------------------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check the security and condition of all external panels on gantry and PC. | Weekly    |

### Collimators

| Good                     | Bad                      | N/A                      | Action                                                              | Frequency |
|--------------------------|--------------------------|--------------------------|---------------------------------------------------------------------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check all collimators for damage to septa and to locking mechanism. | Weekly    |

### COR

| Good                     | Bad                      | N/A                      | Action                                            | Frequency |
|--------------------------|--------------------------|--------------------------|---------------------------------------------------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Check the center of rotation for all collimators. | Weekly    |

### Check of Motion Limits

| Good                     | Bad                      | N/A                      | Action                                                                                 |
|--------------------------|--------------------------|--------------------------|----------------------------------------------------------------------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Verify that the vertical motion axis does not enter into any of the ultimate limits.   |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Verify that the horizontal motion axis does not enter into any of the ultimate limits. |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Verify that the rotational motion axis does not enter into any of the ultimate limits. |

### Lubrication

#### Horizontal Drive

| Good                     | Bad                      | N/A                      | Action                                                                                                      |
|--------------------------|--------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Remove the cover from the gantry base to access the horizontal drive.                                       |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Apply lithium grease with the grease gun on the zerk fitting.                                               |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Apply a small amount of grease to the rails inside the gantry base and to the two gear wheels at the motor. |

#### Vertical Drive

| Good                     | Bad                      | N/A                      | Action                                                                                                                                                          |
|--------------------------|--------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Remove the cover on the side of the Y pillar.                                                                                                                   |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Use the grease gun to apply lithium grease through the zerk fitting on the ball screw nut.                                                                      |
| <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | Remove the rotate gear cover and use the grease gun to apply lithium grease through the two-zerk fittings on the bearing shown in the photo above on the right. |

## System Condition

After performing scheduled PM, an evaluation of the system is performed with the following result (check one):

- ☐ No deficiencies. The image quality test resulted in no differences from required reference values.
- ☐ Slight deficiencies. There is no effect on operation; however they should be corrected preventively. The image quality test resulted in no differences from required reference values.
- ☐ Serious deficiencies. This must be corrected before operation of the system is permitted.

### FSE

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

Name: \_\_\_\_\_

Notes (Additional work to be performed, partial PM, customer requests):

\_\_\_\_\_

\_\_\_\_\_

### Customer

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

Name: \_\_\_\_\_